Lecture 2: Introduction to NLP

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Announcements

- PTEs may be issued during the second week.
  - Register at: https://bit.ly/CS263-PTE
  - For BruinLearn access, email Bryan at bryanzhou008@ucla.edu

- Syllabus updated: https://bit.ly/UCLA-CS263

- You are welcome to audit this course and participate in discussions, even if not enrolled, space permitting.

Course policy

- Grading policy (Tentative):
  - 15% Quizzes (online)
  - 10% Midterm (4/20 in-class)
  - 20% Homework
  - 15% Final Exam (5/11 in-class)
  - 10% Paper Presetnation (peer review, graded by the instructor)
  - 25% Final Project (peer review, graded by the instructor)
  - 5% feedback/participation

Tentative Schedule:

- Week 1:
  - 3/30 Course Introduction
  - 4/1 Deep Learning
- Week 2
  - 4/6 Word Representation; Tokenization
  - 4/8 Neural Language Model; Transformer
- Week 3
  - 4/13 In-context Learning and Prompt Engineering
  - 4/15 Implementation Practice: Running and Training LLMs
- Week 4:
  - 4/20 Midterm. Test-Time Scaling & LLM Reasoning
    - Presentation: [1]
  - 4/22 Post-training Alignment - SFT
    - Presentation: [2]
- Week 5:
  - 4/27 Post-training Alignment - PPO & DPO
    - Presentation: [3]
  - 4/29 Retrieval Augmented Generation
    - Presentation: [4]
- Week 6:
  - 5/4 Quantization; QLora
    - Presentation: [5]
  - 5/6 Evaluation and LLM as Judge
    - Presentation: [6]
- Week 7:
  - 5/11 Final Exam.
  - 5/13 Tool-Use and LLM Agent
    Presentation: [7]
- Week 8:
  - 5/18 Multi-Modal LLM
    - Presentation: [8]
  - 5/20 LLM Safety, Red-teaming,
    - Presentation: [9]
- Week 9
  - 5/25 holiday
  - 5/27 Interpretability, Privacy, and Potential Harms
    - Presentation: [11]
- Week 10
  - 6/1 Final project presentation
  - 6/3 Final project presentation

What you will learn

- How to define an NLP task?
  - Running example – Coreference resolution
- How to design evaluation metrics
- Modeling approaches

Q: [Chris] = [Mr. Robin] ?

Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh.** As **a boy**, **Chris** lived in a pretty home called **Cotchfield Farm**. When **Chris** was three years old, **his father** wrote a poem about **him**. The poem was printed in a magazine for others to read. **Mr. Robin** then wrote a book

P.S. In fact, Alan Alexander Milne is the author.

Slide modified from Dan Roth

Complex Decision Structure

Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh.** As **a boy**, **Chris** lived in a pretty home called **Cotchfield Farm**. When **Chris** was three years old, **his father** wrote a poem about **him**. The poem was printed in a magazine for others to read. **Mr. Robin** then wrote a book

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Co-reference Resolution

Identify all mentions that refer to the same entity

Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh**. As a **boy**, **Chris** lived in a pretty home called **Cotchfield Farm**. When **Chris** was three years old, **his father** wrote a poem about **him**. The poem was printed in a magazine for others to read. **Mr. Robin** then wrote a book

How to define this task?

Co-reference Resolution

Input: A piece of text

Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh.** As **a boy**, **Chris** lived in a pretty home called **Cotchfield Farm**. When **Chris** was three years old, **his father** wrote a poem about **him**. The poem was printed in a magazine for others to read. **Mr. Robin** then wrote a book

Output: Clusters of phrases

Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh**. As a **boy**, **Chris** lived in a pretty home called **Cotchfield Farm**. When **Chris** was three years old, **his father** wrote a poem about **him**. The poem was printed in a magazine for others to read. **Mr. Robin** then wrote a book

Co-reference Resolution

Task 1: Mention detection

Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh**. As a **boy**, **Chris** lived in a pretty home called **Cotchfield Farm.** When **Chris** was three years old, **his father** wrote a poem about **him**. The **poem** was printed in a **magazine** for others to read. **Mr. Robin** then wrote a book

Co-reference Resolution

Task 1: Mention detection

Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh**. As a **boy**, **Chris** lived in a pretty home called **Cotchfield Farm**. When **Chris** was three years old, **his father** wrote a poem about **him**. The **poem** was printed in a **magazine** for others to read. **Mr. Robin** then wrote a book

Task 2: Cluster the mentions

Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh**. As a **boy**, **Chris** lived in a pretty home called **Cotchfield Farm**. When **Chris** was three years old, **his father** wrote a poem about **him**. The poem was printed in a magazine for others to read. **Mr. Robin** then wrote a book

Mentions

- Span of words that refer to one or few entities
- Mentions can be
  - Pronoun (he, she, they)
  - Common noun (boy, girl, dog)
  - Named Entity (Chris, Google)
- Not every noun/pronoun is mention
  - It is raining outside
  - No student wants to take the final test

Mentions

- Mention can be nested

"[I] voted for [Nader] because [he] was most aligned with [[my] values]," [she] said

- Define mention boundary can be tricky

At this moment, the US President Biden , who is far away in Europe , may have more than enough energy.

v.s.

At this moment, the US President Biden , who is far away in Europe , may have more than enough energy.

Resolving coreference

- Sometimes can be done by leveraging grammar rules

The secretary called the physician and told him about a new patient

Resolving coreference

- Sometimes require commonsense

The umbrella cannot fit into the bin, because it is too small

v.s.

The umbrella cannot fit into the bin, because it is too big

How to evaluate?

❖ Ground Truth (annotated by linguistics)

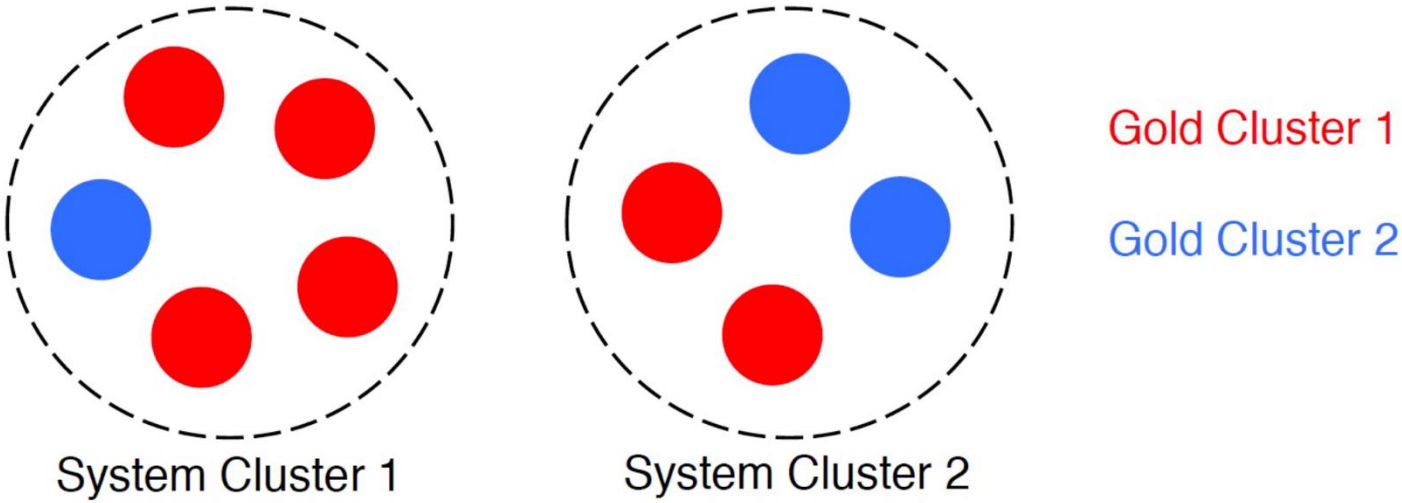
Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh**. As a **boy**, **Chris** lived in a pretty home called **Cotchfield Farm**. When **Chris** was three years old, **his father** wrote a poem about **him**. The poem was printed in a magazine for others to read. **Mr. Robin** then wrote a book

❖ Model prediction

Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh**. As a **boy**, **Chris** lived in a pretty home called **Cotchfield Farm**. When **Chris** was three years old, **his father** wrote a poem about **him**. The poem was printed in a magazine for others to read. **Mr. Robin** then wrote a book

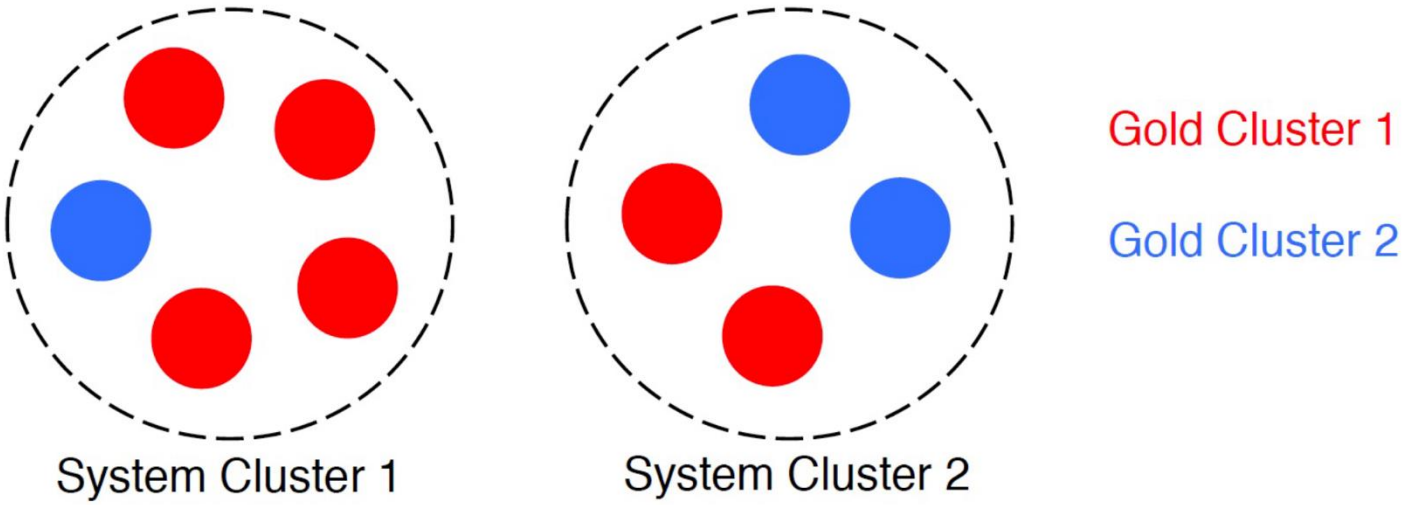
How to evaluate the performance?

- Can be viewed as a supervised clustering task
- Evaluating cluster performance is tricky



Metric 1: Exact Matching Accuracy

- The system gets credit only when its answer exactly match the ground truth



	Ground Truth (Gold annotation)	System output (prediction)	score
Doc 1	**Christopher Robin** is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh**. As a **boy**, **Chris** lived in a pretty home called **Cotchfield Farm**.	**Christopher Robin** is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh**. As a **boy**, **Chris** lived in a pretty home called **Cotchfield Farm**,	100%
	⋮	⋮	⋮
Doc K	**[Bill Clinton]**, recently elected as the **[President of the USA]**, has been invited by the **[Russian President]**, **[Vladimir Putin]**, to visit **[Russia]**. **[President Clinton]** said that **[he]** looks forward to strengthening ties between **[USA]** and **[Russia]**.	**[Bill Clinton]**, recently elected as the **[President of the USA],** has been invited by the **[Russian President]**, **[Vladimir Putin]**, to visit **[Russia]**. **[President Clinton]** said that **[he]** looks forward to strengthening ties between **[USA]** and **[Russia]**.	0%

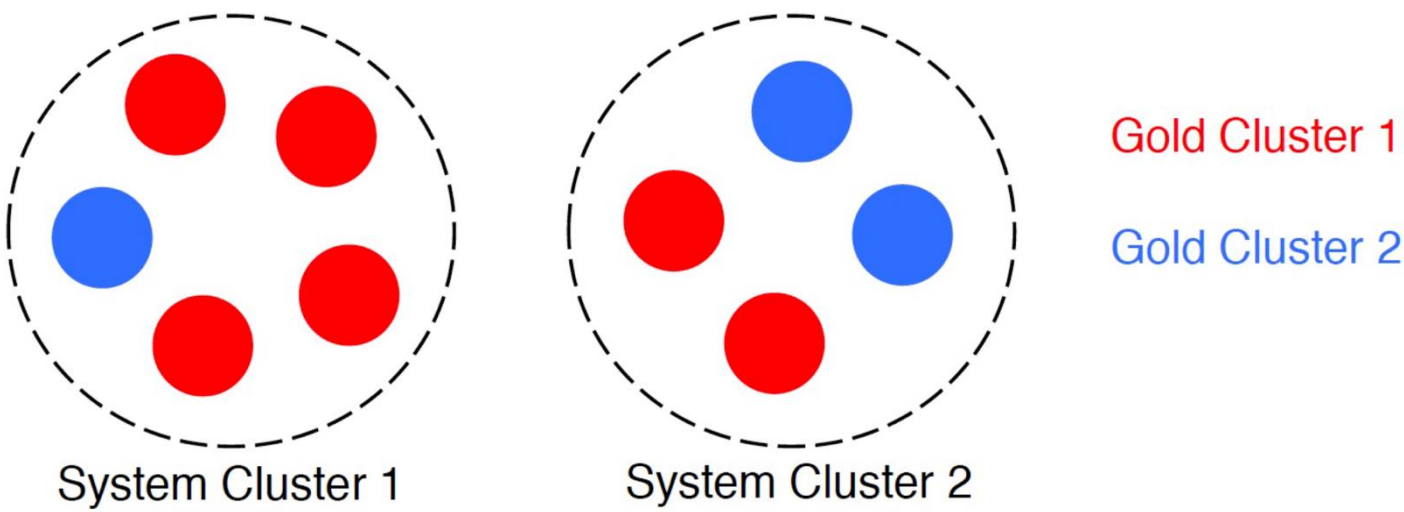
Average accuracy 50%

MUC evaluation metric

- Compute precision/recall of mention links

Ground Truth

Model Prediction	Positive	Negative
Positive	True Positive (TP)	False Positive (FP)
Negative	False Negative (FN)	True Negative (TN)



MUC evaluation metric

- How many True Positive pairs?

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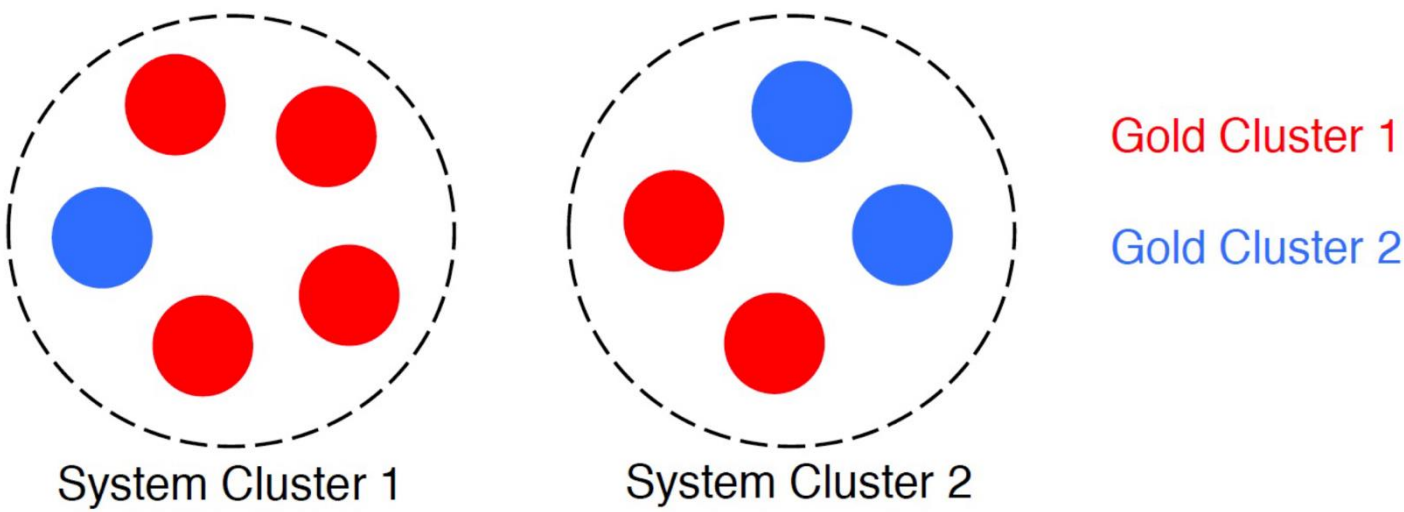
LCKW

Model Prediction \ Ground Truth	Positive	Negative
Positive	True Positive (TP)	False Positive (FP)
Negative	False Negative (FN)	True Negative (TN)



MUC evaluation metric

- How many Total (TP+FP) pairs?

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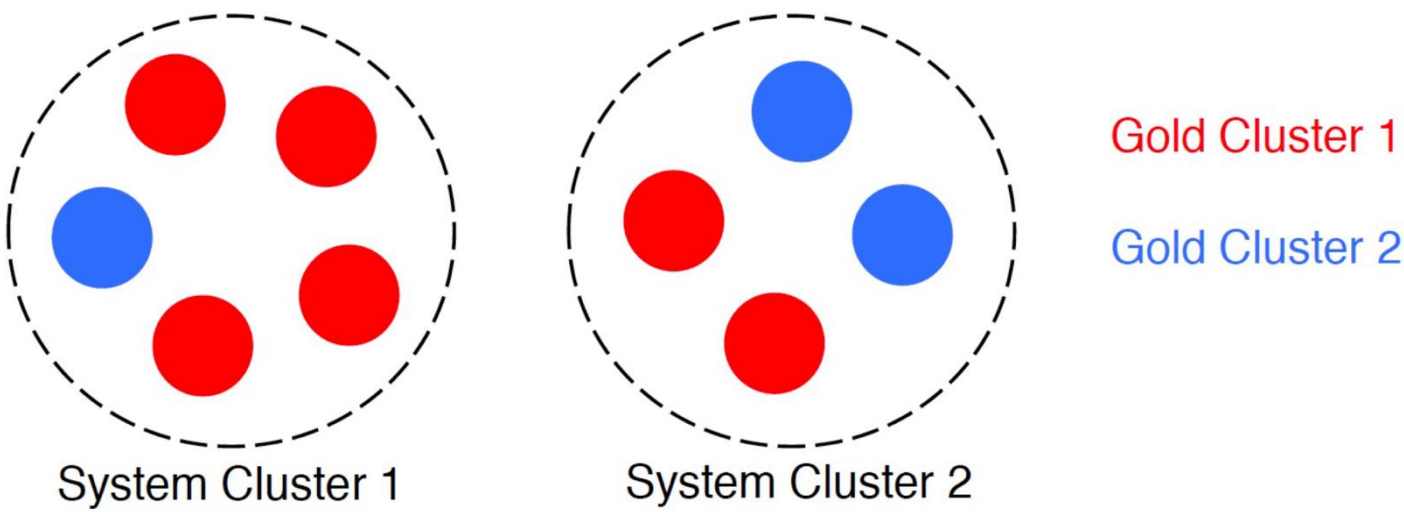
Model Prediction \ Ground Truth	Positive	Negative
Positive	True Positive (TP)	False Positive (FP)
Negative	False Negative (FN)	True Negative (TN)



MUC evaluation metric

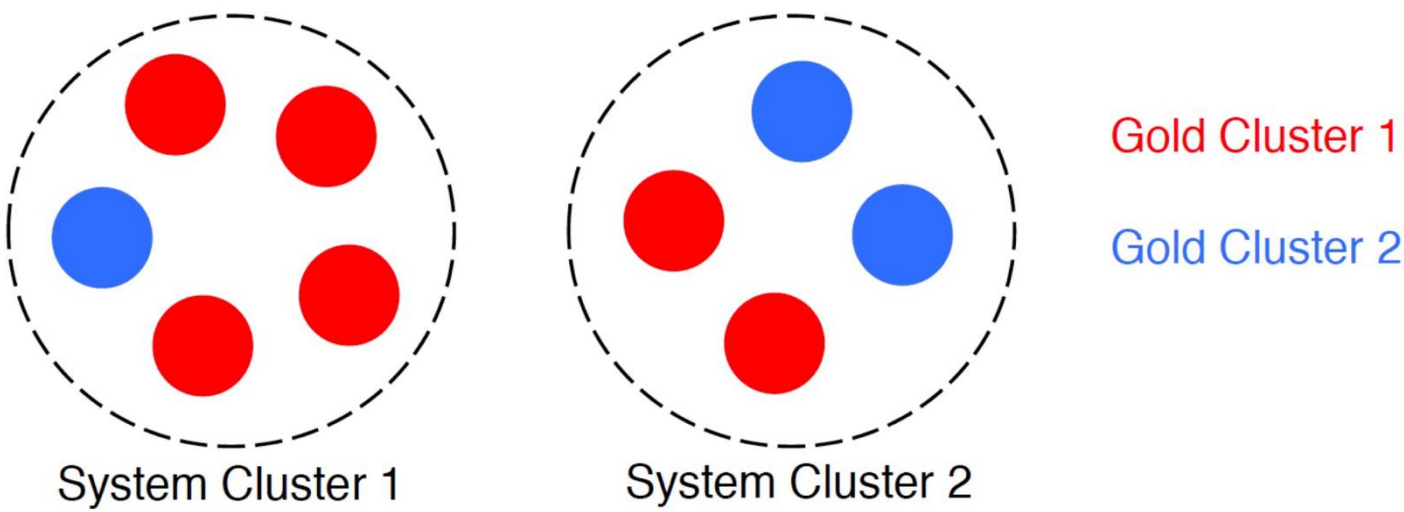
- Compute precision/recall of mention links
- Precision:

True Positive (TP)

$$\frac{\text{\# predicted pairs refer to the same entity}}{\text{\#predicted mention pairs}}$$

True Positive (TP) + False Positive (FP)

8/16



MUC evaluation metric

- How many False Negative pairs?

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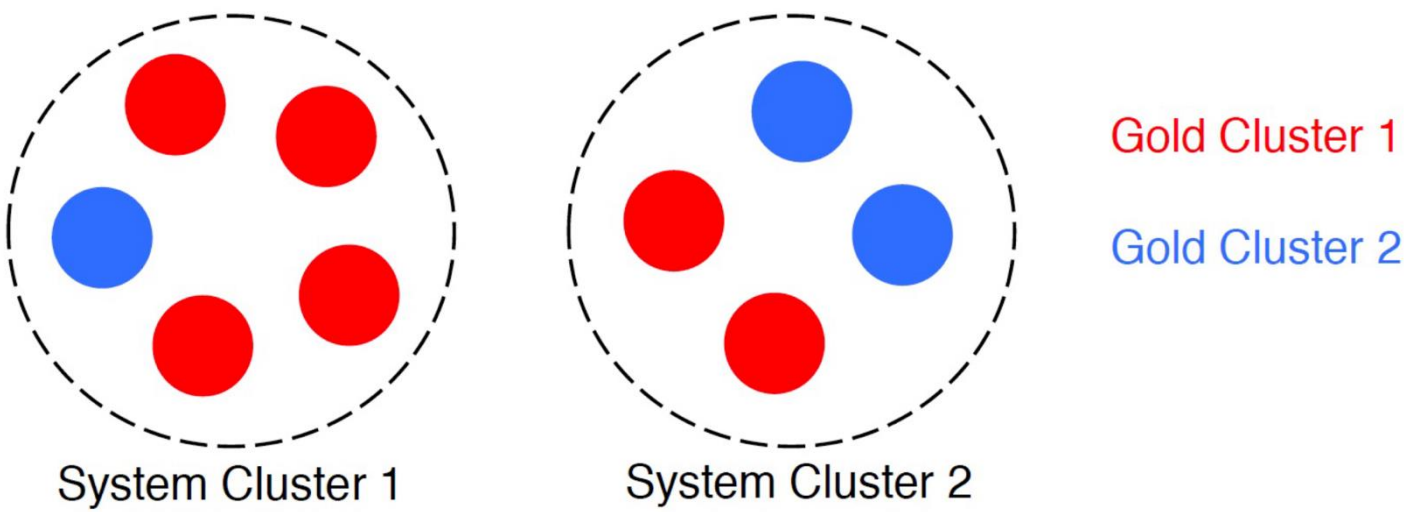
Ground Truth

Model Prediction	Positive	Negative
Positive	True Positive (TP)	False Positive (FP)
Negative	False Negative (FN)	True Negative (TN)



MUC evaluation metric

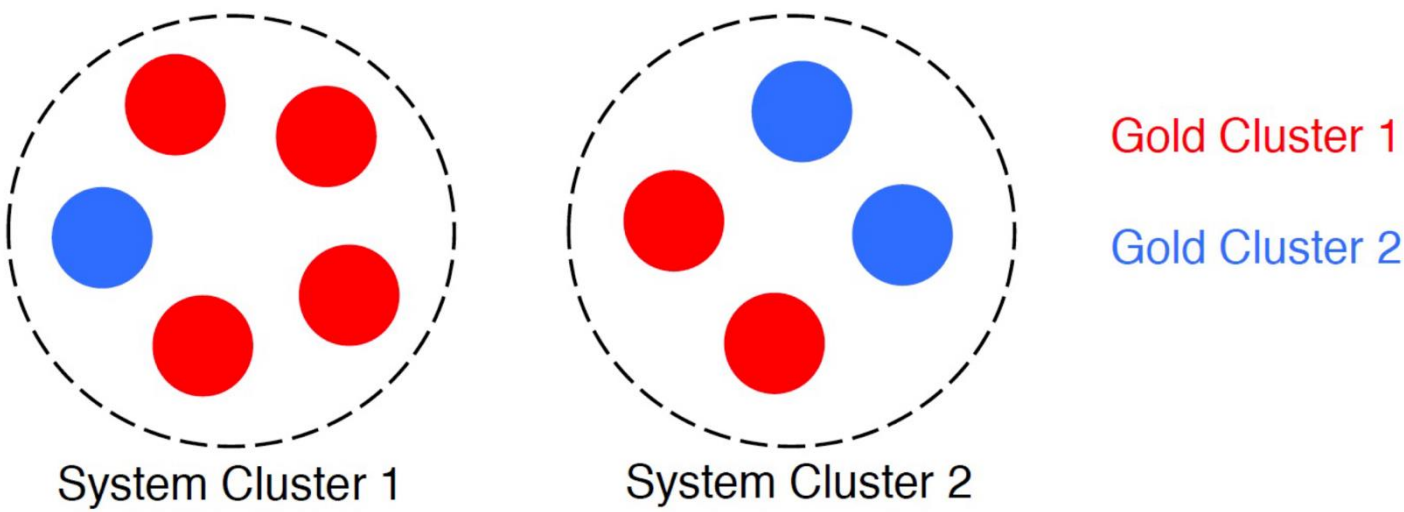
- Compute precision/recall of mention links
- Recall:

True Positive (TP)

$$\frac{\text{\# gold pairs in the same predicted } \textit{clusters}}{\text{\#gold mention pairs}}$$

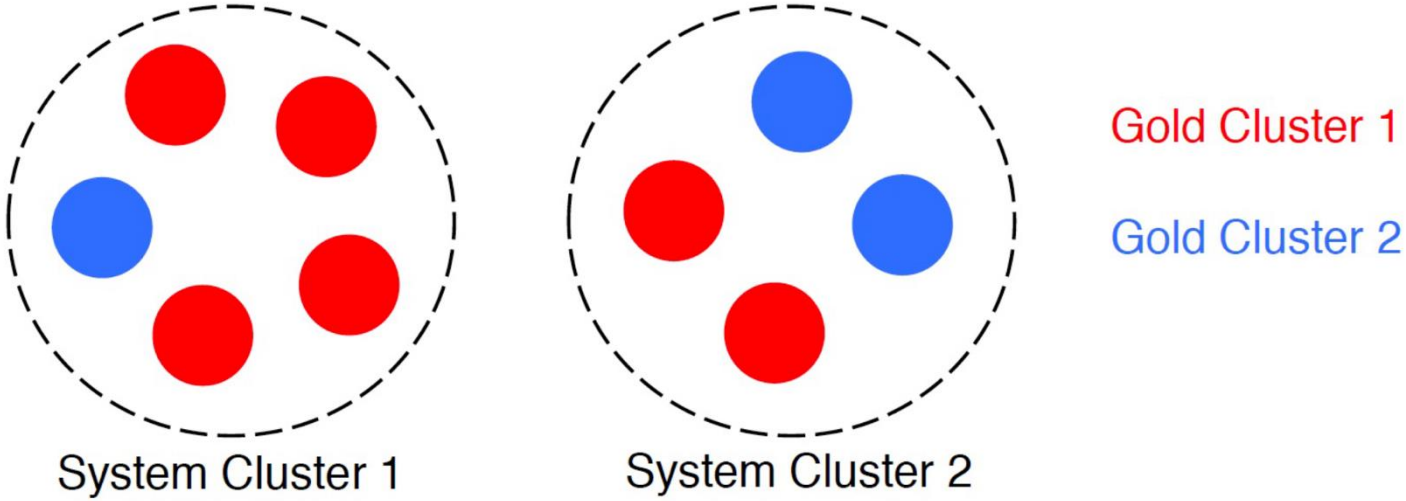
True Positive (TP) + False Negative (FN)

8/18



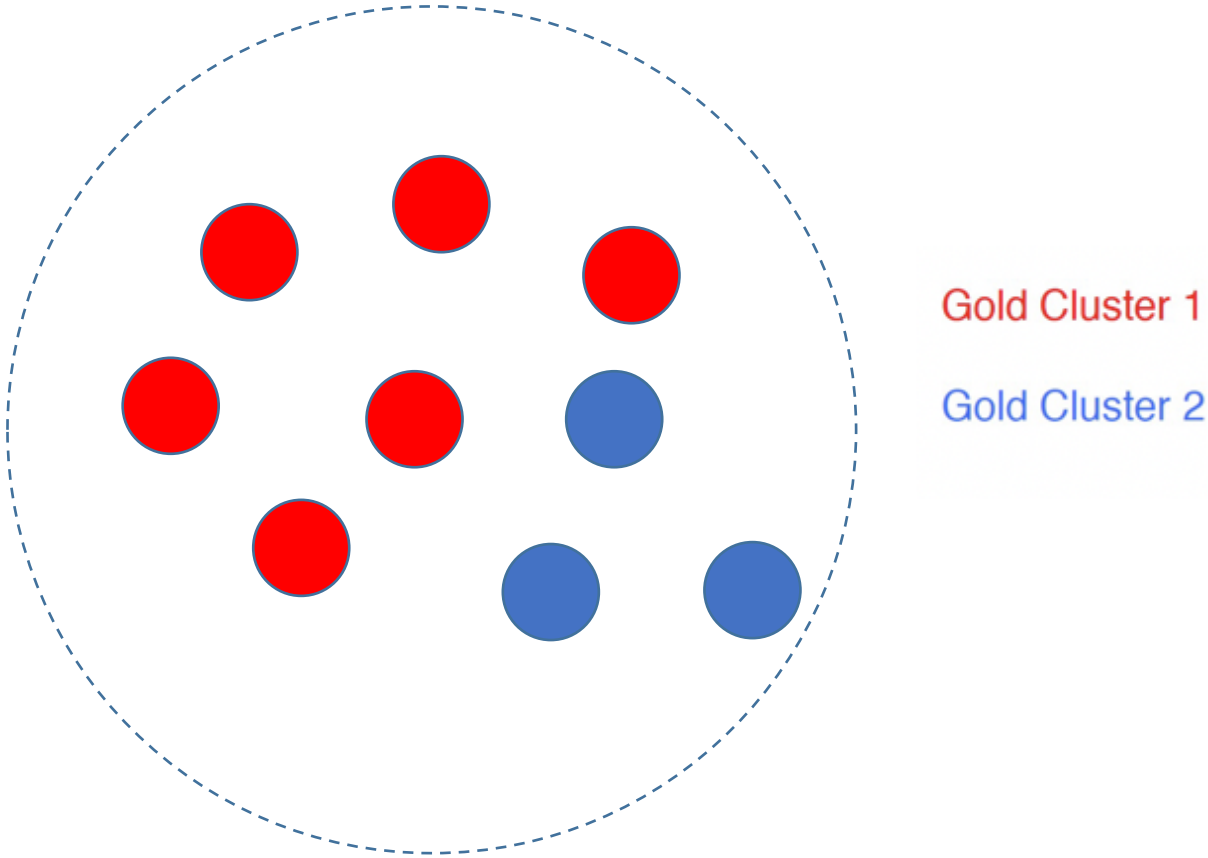
MUC evaluation metric

- Increase precision will decrease recall
- **F1 = harmonic mean of precision and recall**
  = 2 / (1/precision + 1/ recall)
  = 0.47



Your turn

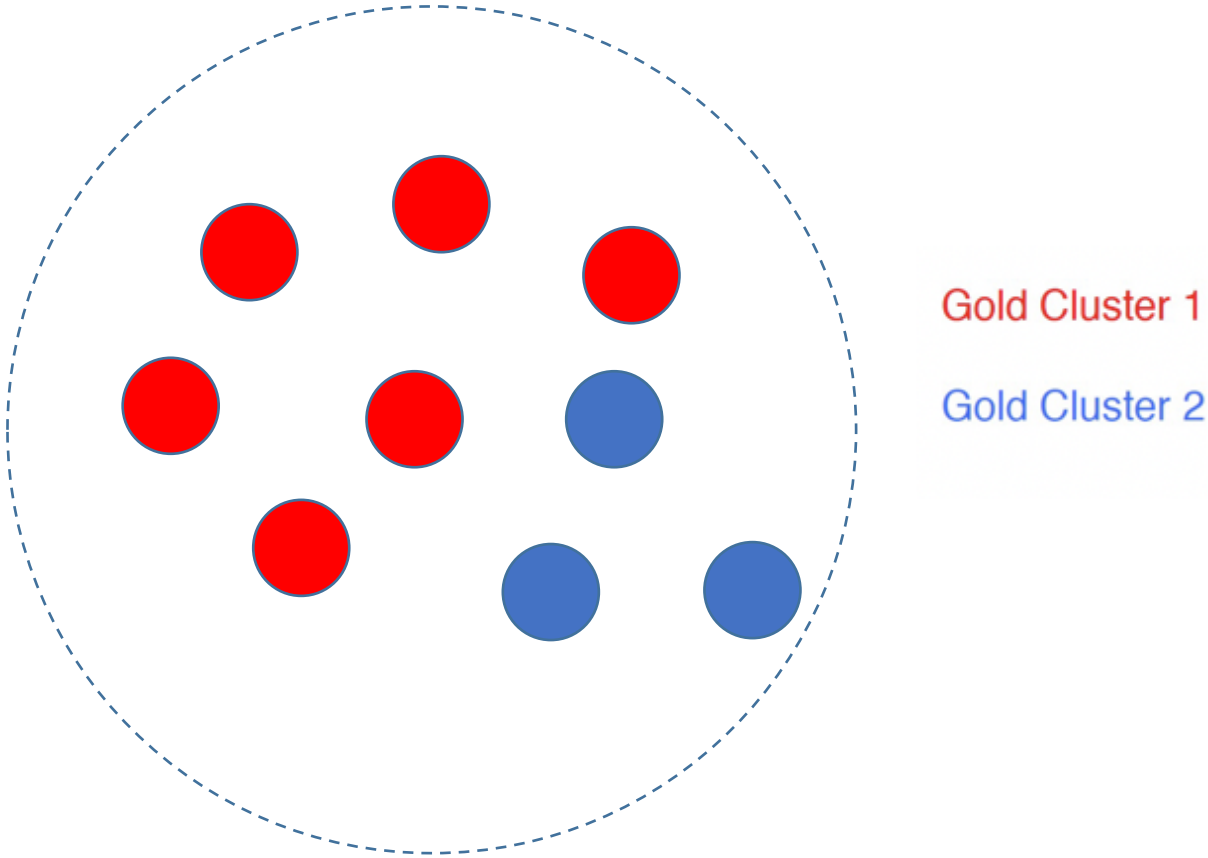
- What is the Recall score of the following system prediction



Issue of MUC score

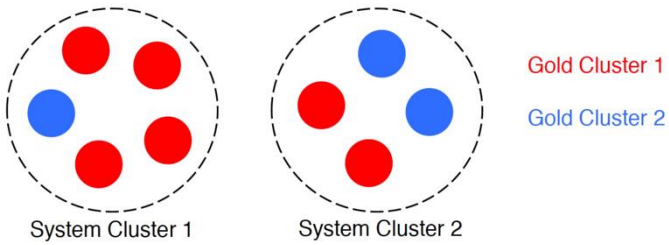
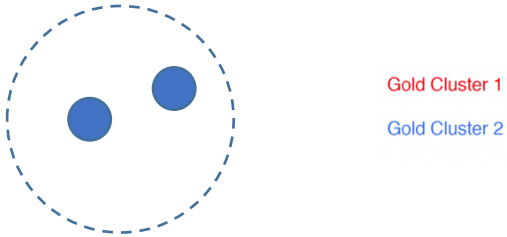
- What is the Recall of the following system prediction

Recall = 1



How to aggregate F1 score

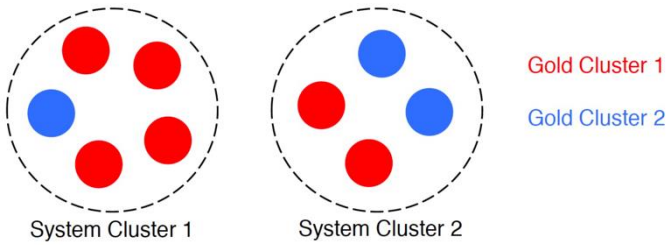
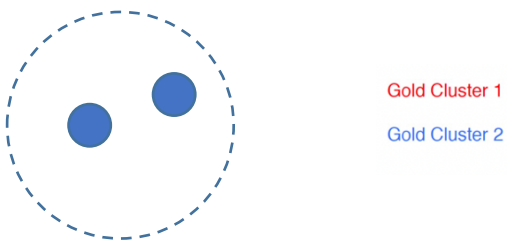
- Macro F1

		Precision	Recall	F1
Doc 1		8/16	8/18	0.47
Doc 2		1/1	1/1	1/1

Macro F1 = Avg (F1) = (0.47+1)/2 = 0.74

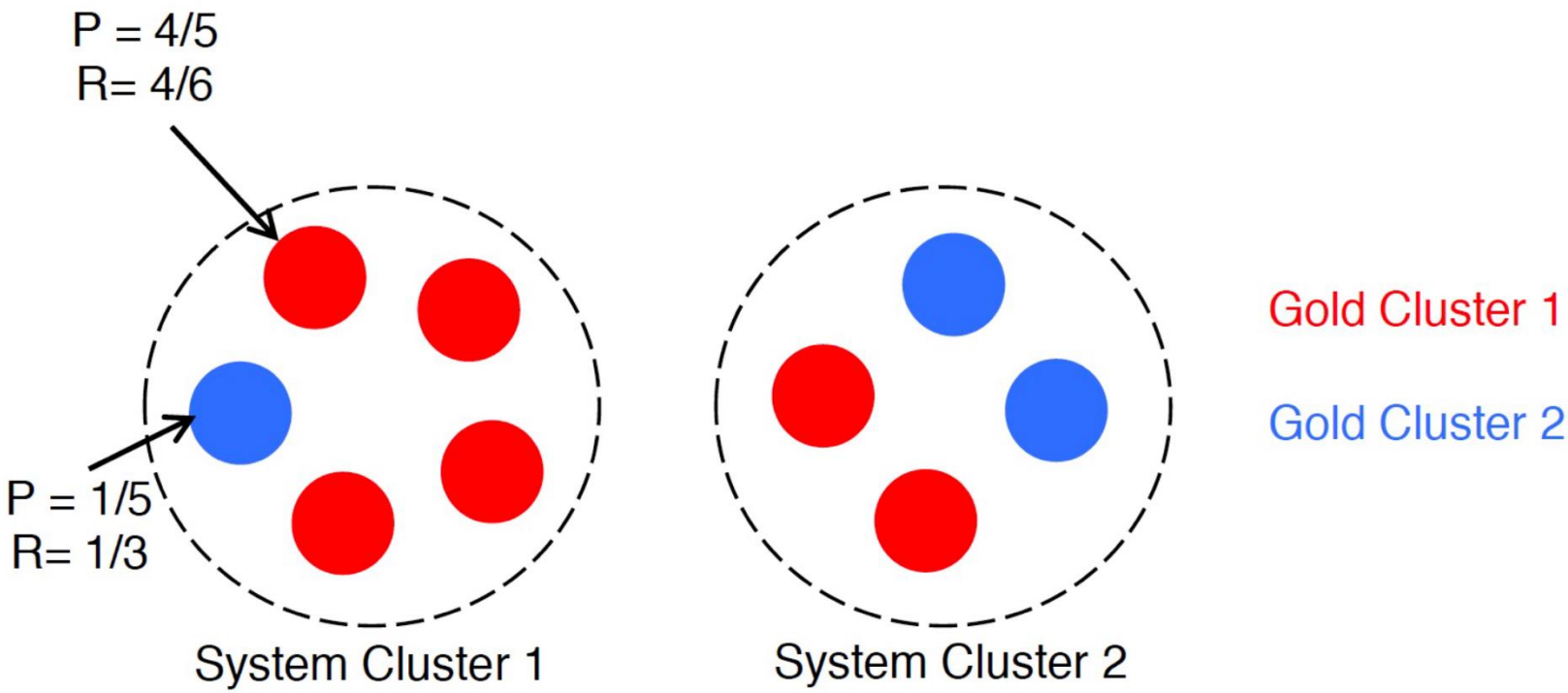
How to aggregate F1 score

- Micro F1

		Precision	Recall	F1
Doc 1		8/16	8/18	0.47
Doc 2		1/1	1/1	1/1
Aggregate FP, FN, TP, TN		9/17	9/19	0.5

BCUB evaluation metric

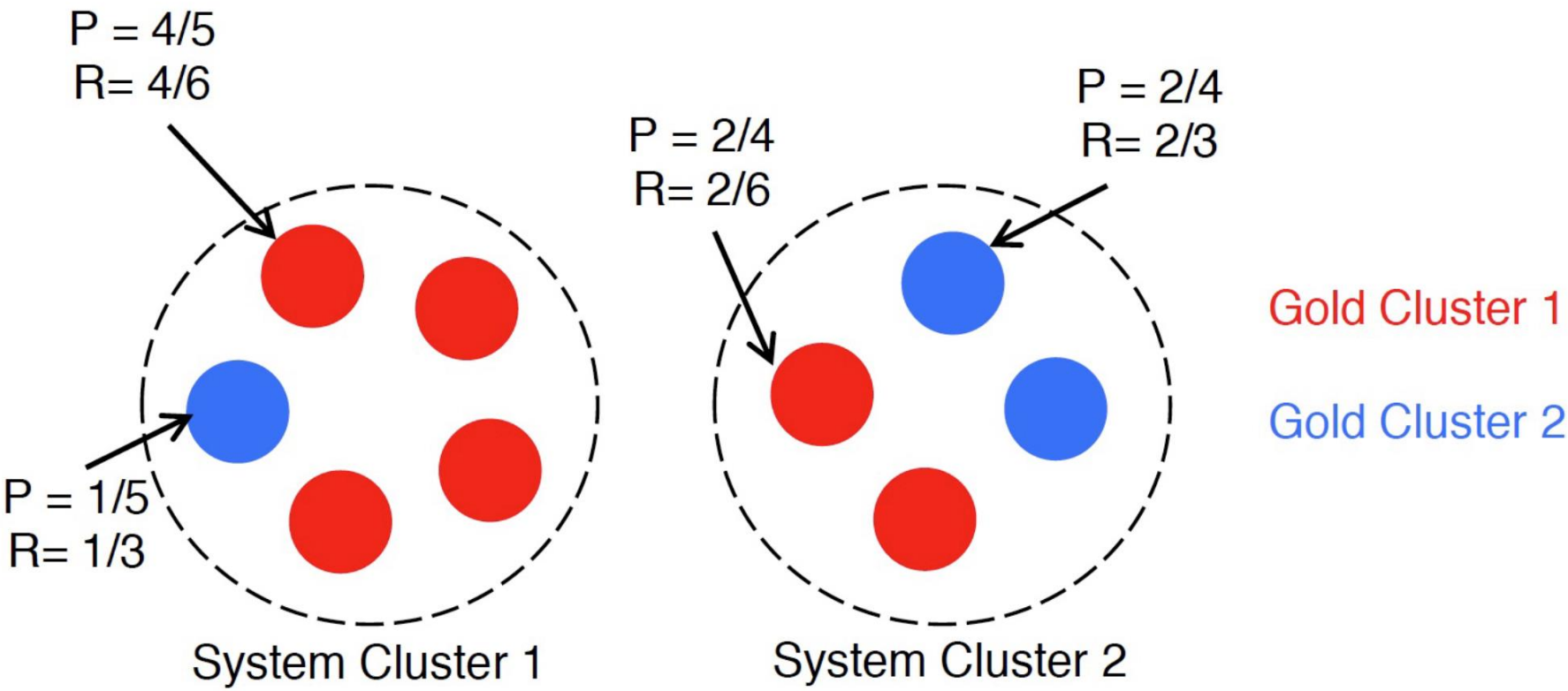
- Compute precision/recall of each mention



BCUB evaluation metric

- Average the individual Ps & Rs

$$P = [4(4/5) + 1(1/5) + 2(2/4) + 2(2/4)] / 9 = 0.6$$



Metrics might not be consistent

[The American administration]$_{(1)}$ committed a fatal mistake when [it$_1$]$_{(1)}$ [executed]$_{(2)}$ [this man]$_{(3)}$, in a way for which [it$_2$]$_{(1)}$ will pay a hefty price in the near future. [[His$_1$]$_{(3)}$ survival]$_{(4)}$ would have benefited [it$_3$]$_{(1)}$ much more than [[his$_2$]$_{(3)}$ execution]$_{(2)}$ if [they$_1$]$_{(1)}$ understood politics as [they$_2$]$_{(1)}$ should, because [[his$_3$]$_{(3)}$ survival]$_{(4)}$ could have been a card to threaten [the sectarians]$_{(5)}$ and keep [them$_1$]$_{(5)}$ as servants to [them$_1$]$_{(1)}$ and [their]$_{(1)}$ schemes.

	Response entities
cr_1	r_1={the American administration, it$_1$, it$_2$, it$_3$} , r_2={they$_1$, they$_2$, them, their}
cr_2	r_1={the American administration, it$_1$, it$_2$, it$_3$}

	MUC	B^3	$CEAF_m$	$CEAF_e$	BLANC
cr_1	92.30	66.66	50.00	44.44	60.00
cr_2	60.00	40.00	66.66	66.66	32.29

Coreference Models

Brainstorm -- How to deal with this problem?

Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh**. As a **boy**, **Chris** lived in a pretty home called **Cotchfield Farm**. When **Chris** was three years old, **his father** wrote a poem about **him**. The poem was printed in a magazine for others to read. **Mr. Robin** then wrote a book

Co-reference Resolution

- Learn a pairwise similarity measure (local predictor)
- Key questions:
  - How to identify mention span
  - How to learn the similarity function
  - How to cluster mentions

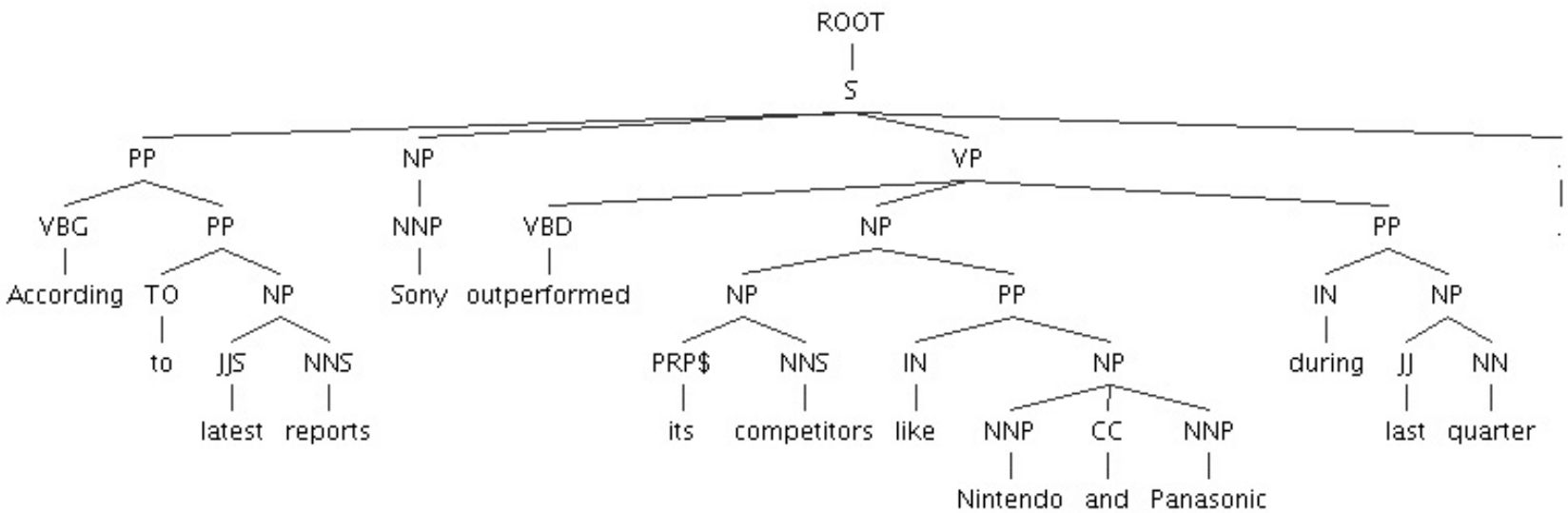
Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh**. As a **boy**, **Chris** lived in a pretty home called **Cotchfield Farm**. When **Chris** was three years old, **his father** wrote a poem about **him**. The poem was printed in a magazine for others to read. **Mr. Robin** then wrote a book

Mention Span Detection

- Earlier pipeline approach – build a mention detector
  - Include pronouns, name entities, common nouns (using information from parse tree) (1994--2017)
- End-to-End Approach
  - Learning a detection model jointly with coreference models by neural network (2017-)
  - Seq-2-Seq Models

Mention-Span Detection

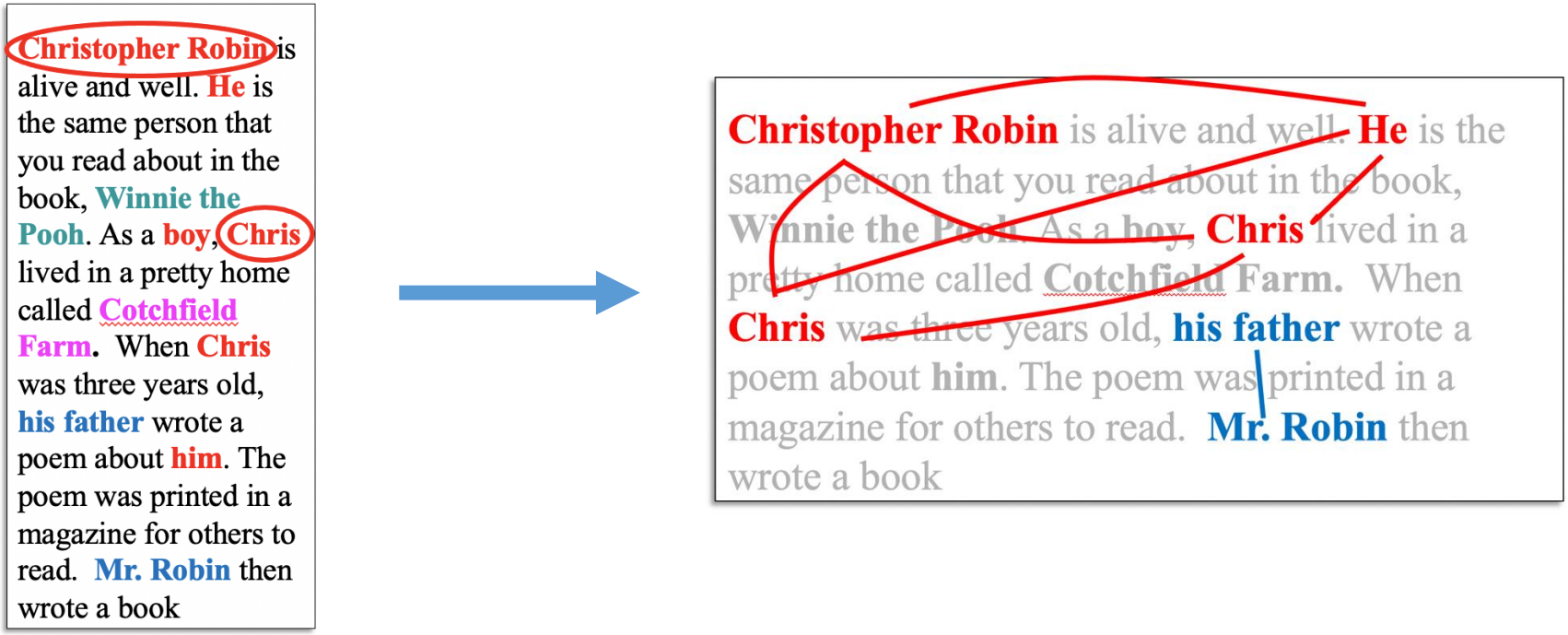
Constituency Parsing: break down a sentences into phrases (a.k.a. constituency)



According to latest reports Sony outperformed its competitors like Nintendo and Panasonic during last quarter.

Pairwise Coreference Models

1. Compute similarity between word spans
2. Generate coreference clusters



How to learn similarity function

Feature-based approach, learn a classifier with features:

What features we can use?

Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh**. As a **boy**, **Chris** lived in a pretty home called **Cotchfield Farm.** When **Chris** was three years old, **his father** wrote a poem about **him**. The poem was printed in a magazine for others to read. **Mr. Robin** then wrote a book

How to learn similarity function

Feature-based approach, learn a classifier with features:

Lexical Features

Christopher, Chris

Number agreement

IBM announced a new machine translation product yesterday. *They* have been working on it for 20 years.

Gender or Noun Class Agreement:

Maryam has a theorem. She is exciting. (she=Maryam, not the theorem)

Maryam has a theorem. It is exciting. (it=the theorem, not Maryam)

How to encode Lexical Features?

- Naïve way: represent words as atomic symbols: student, talk, university
- Represent word as a "one-hot" vector

  [0 0 0 1 0 … 0]

  egg student talk university happy buy
- How large is this vector?
  - Vector dimension = number of words in vocabulary

    PTB data: ~50k, Google 1T data: 13M
- $v \cdot u = ?$

How to encode Lexical Features?

- Word embedding

$$expect = \begin{pmatrix} 0.286 \\ 0.792 \\ -0.177 \\ -0.107 \\ 0.109 \\ -0.542 \\ 0.349 \\ 0.271 \\ 0.487 \end{pmatrix}$$

- Distance between v and u can be define as similarity(v,u) = $\frac{v \cdot u}{||v||\ ||u||}$ (cosine similarity)

Training/Inference with Coreference Models

How to learn the model?

As a **boy**, **Chris**$_1$ lived in a pretty home called **Cotchfield Farm.** When **Chris**$_2$ was three years old, **his father**$_3$ wrote a poem about **him**$_4$. The poem was printed in a magazine for others to read. **Mr. Robin**$_5$ then wrote a book

Positive Samples
(Chris$_1$, him$_4$)
(Chris$_2$, him$_4$)
(Chris$_1$, Chris$_2$)
(his father$_3$, Mr. Robin$_5$)

Negative Samples
(Chris$_1$, his father$_3$)
(Chris$_2$, his father$_3$)
(him$_4$, his father$_3$)
(Chris$_1$, Mr. Robin$_5$)
(Chris$_2$, Mr. Robin$_5$)
(him$_4$, Mr. Robin$_5$)

Train a binary classification (e.g., logistic regression) model based on features

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Computer Science

Option 1: logistic regression
Predict P(y = 1 | **x**)

Input: $x \in \mathbb{R}^d$

Output: $y \in \{1, -1\}$

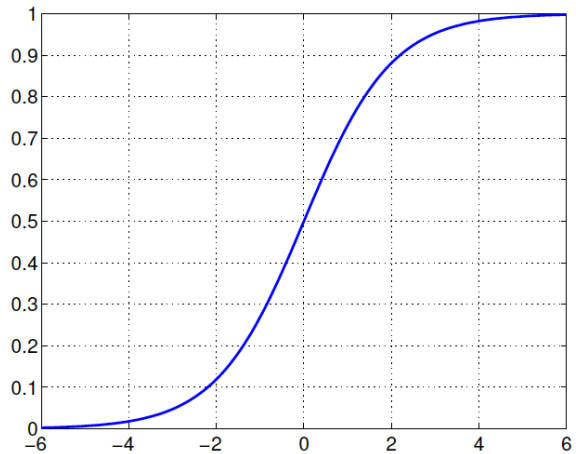
Build a model $h(x)$ such that

$$h(x) = \sigma(w^T x + b) \approx P(y = 1|x)$$

a regression problem

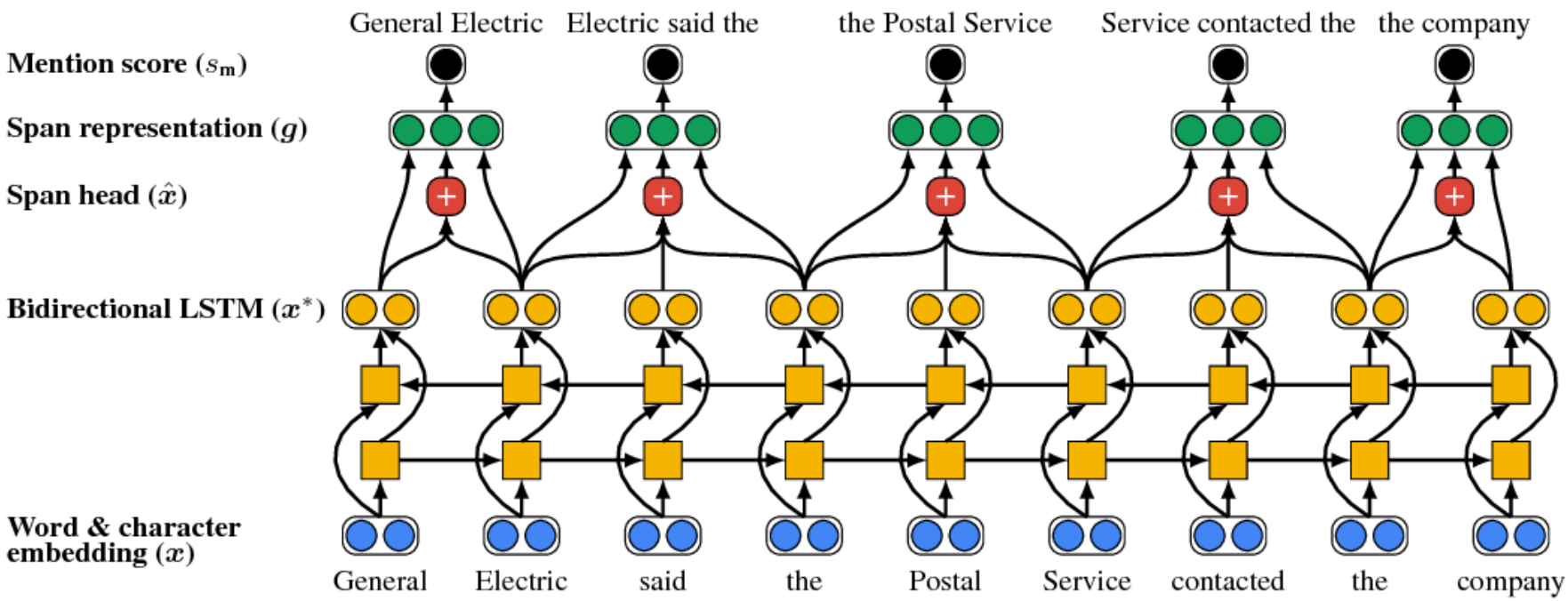
σ is a sigmoid function

$$\sigma(z) = \frac{\exp(z)}{1 + \exp(z)}$$
$$= \frac{1}{1+\exp(-z)}$$



Option 2: Neural Network

- With neural network, the model can be trained end-to-end



How to generate clustering?

Assume we have a model that can tell whether two phrases refer to the same entity or not. How can we use it to generate coreference clusters?

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Greedy (Left-Linking Tree)

Greedy Best-Left-Link Clustering

[Bill Clinton], recently elected as the **[President of the USA]**, has been invited by the [Russian President], [Vladimir Putin], to visit [Russia]. [President Clinton] said that [he] looks forward to strengthening ties between [USA] and [Russia].

Greedy Best-Left-Link Clustering

[Bill Clinton], recently elected as the **[President of the USA]**, has been invited by the **[Russian President]**, **[Vladimir Putin], to visit [Russia]. [President Clinton] said that [he] looks forward to strengthening** ties between **[USA]** and **[Russia]**.

Greedy Best-Left-Link Clustering

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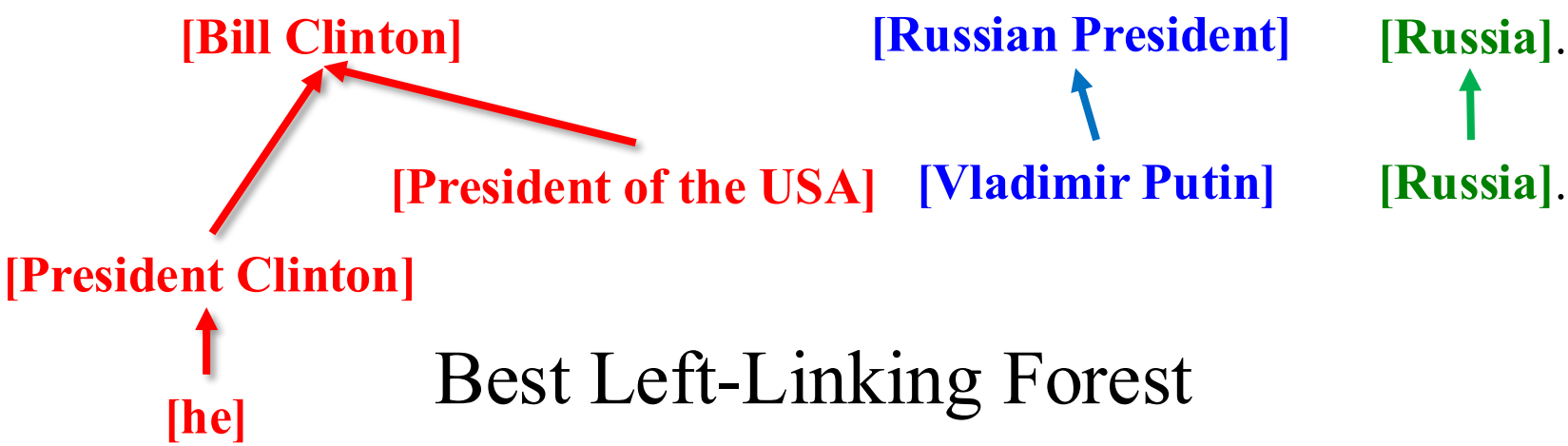
Greedy Best-Left-Link Clustering

[Soon+ 01, Bengtson+ 08]

[Bill Clinton], recently elected as the **[President of the USA]**, has been invited by the **[Russian President]**, **[Vladimir Putin]**, to visit **[Russia]**. **[President Clinton]** said that **[he]** looks forward to strengthening ties between **[USA]** and **[Russia]**.



Best Left-Linking Forest

Seq2-Seq for Corefernece

Input: *Speaker-A* I still have n't gone to that fresh French restaurant by your house
Prediction: SHIFT: next sentence

Input: *Speaker-A* I_2 still have n't gone to that fresh French restaurant by your house *Speaker-A* I_{17} 'm like dying to go there
Prediction:
A $I_{17} \rightarrow I_2$
B SHIFT: next sentence

Input: *Speaker-A* [1 I] still have n't gone to that fresh French restaurant by your house *Speaker-A* [1 I] 'm like dying to go there *Speaker-B* You mean the one right next to the apartment
Prediction:
A You → [1
B the apartment → your house
C the one right next to the apartment → that fresh French restaurant by your house
D SHIFT: next sentence

Input: *Speaker-A* [1 I] still have n't gone to [3 that fresh French restaurant by [2 your house]] *Speaker-A* [1 I] 'm like dying to go there *Speaker-B* [1 You] mean [3 the one right next to [2 the apartment]] *Speaker-B* yeah yeah yeah
Prediction: SHIFT: next sentence

- Coreference Resolution through a seq2seq Transition-Based System

  Bernd Bohnet, Chris Alberti, Michael Collins, TACL 22

ChatGPT

KW

Identify all the phrases that refer to the same entity and mark them with [ ].

Christopher Robin is alive and well. He is the same person that you read about in the book, Winnie the Pooh. As a boy, Chris lived in a pretty home called Cotchfield Farm. When Chris was three years old, his father wrote a poem about him. The poem was printed in a magazine for others to read. Mr. Robin then wrote a book

[Christopher Robin] is alive and well. [He] is the same person that you read about in the book, Winnie the Pooh. As a boy, [Chris] lived in a pretty home called Cotchfield Farm. When [Chris] was three years old, his father wrote a poem about [him]. The poem was printed in a magazine for others to read. Mr. Robin then wrote a book.

Identify all the phrases (called mentions) that refer to the same entity and cluster them into equivalent classes. Make sure also to mark nested mentions.

Example input: The secretary called the physician and told him about a new patient

Example output: [The secretary]_1 called [the physician]_2 and told [him]_2 about [a new patient]_3

--

Christopher Robin is alive and well. He is the same person that you read about in the book, Winnie the Pooh. As a boy, Chris lived in a pretty home called Cotchfield Farm. When Chris was three years old, his father wrote a poem about him. The poem was printed in a magazine for others to read. Mr. Robin then wrote a book

I'm mapping all mentions into coreference clusters while preserving nested mentions, especially appositives and inner names or possessors. One ambiguous spot is "Mr. Robin," which in context links back to Christopher Robin rather than introducing a new father entity.

On a closer pass, I'm revising that call: "Mr. Robin" fits better with his father, and that also makes the final "a book" co-refer with "the book, Winnie the Pooh."

Thought for 27m 28s ›

Assuming **"Mr. Robin" refers to Chris's father**, the clustering is:

```
[Christopher Robin]_1 is alive and well. [He]_1 is [[the same person]_1 that [you]_
```

Equivalent classes:

1 = Christopher Robin, He, the same person, a boy, Chris, Chris, his, him
2 = you
3 = the book, Winnie the Pooh, a book
4 = a pretty home, Cotchfield Farm
5 = his father, Mr. Robin
6 = a poem, The poem
7 = a magazine
8 = others

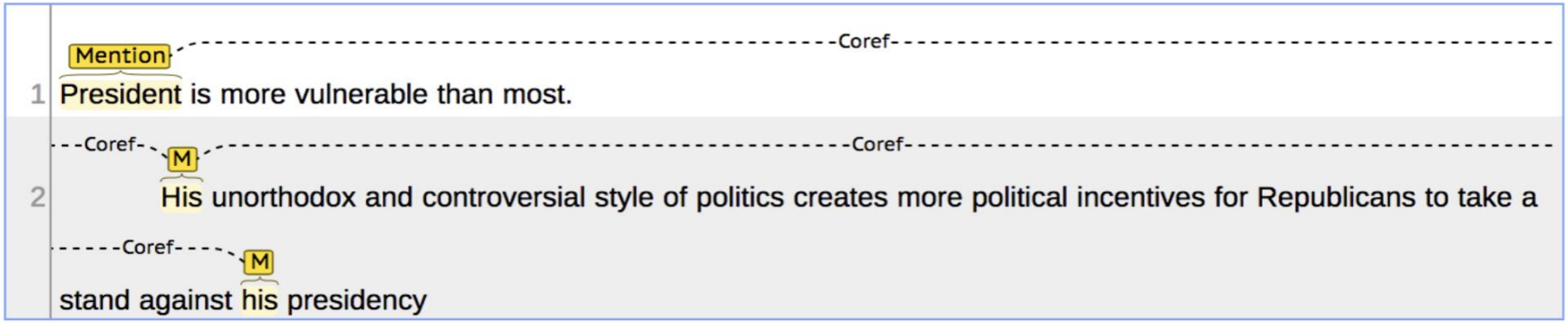
The only real ambiguity is **"Mr. Robin"**; I linked it to **his father** because that fits the local discourse best.

Biases in Coreference Resolution

Gender Bias in Coreference Resolution: Evaluation and Debiasing Methods

Jieyu Zhao, Tianlu Wang, Mark Yatskar, Vicente Ordonez, and Kai-Wei Chang, in NAACL (*short*), 2018.

Model fails for female when given same context



his ⇒ her

[1]Zhao et al. Gender Bias in Coreference Resolution: Evaluation and Debiasing Methods. NAACL 2018.
[2]Rudinger et al. Gender Bias in Coreference Resolution. NAACL 2018

Wino-bias data

- Stereotypical dataset

The physician hired the secretary because he was overwhelmed with clients.

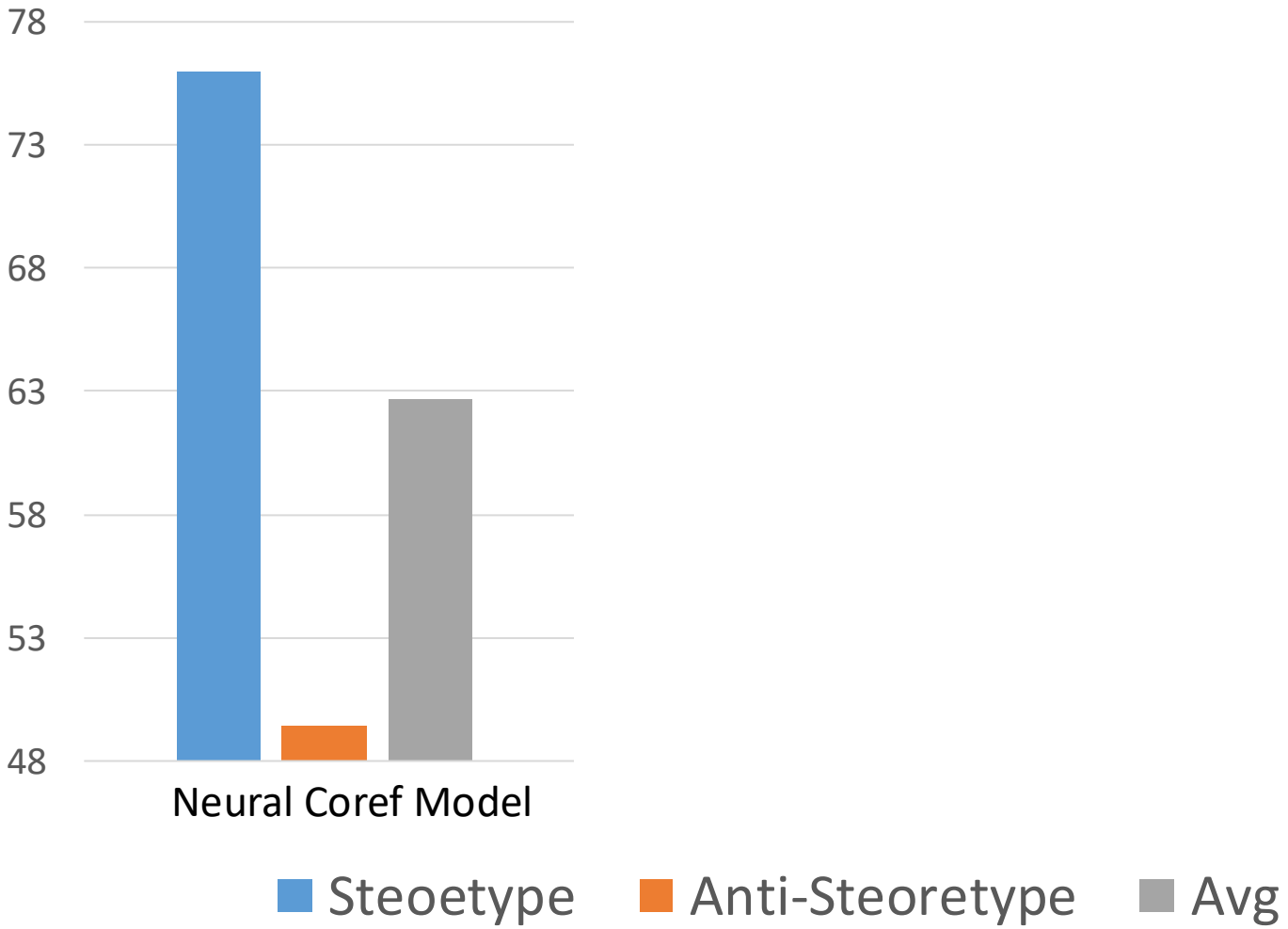
The physician hired the secretary because she was highly recommended.

- Anti-stereotypical dataset

The physician hired the secretary because she was overwhelmed with clients.

The physician hired the secretary because he was highly recommended.

Gender bias in Coref System



The doctor hired the secretary because he is highly organized and efficient. What is he referring to?

Thought for a second

"He" refers to **the secretary**. The qualities "highly organized and efficient" describe the person the doctor hired.

Continue the sentence

The doctor hired the secretary because he is

The doctor hired the secretary because he is looking to expand his practice.

Continue the sentence

The doctor hired the secretary because she is

The doctor hired the secretary because she is highly organized and efficient.